

Impact of Local Time on march 17th 2015 Geomagnetic Storm response at mid-low latitudes

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Abstract

This study analyzes the impact of the partial ring current and Prompt Penetration Electric Fields (PPEFs) on the local geomagnetic response, using data from four mid-to-low latitude observatories separated by approximately 60°–90° in longitude, covering different local time sectors. We focus on the geomagnetic storm of 17 March 2015. Using Fourier analysis, we reconstruct the magnetic signatures of ionospheric currents. Our results demonstrate that the local geomagnetic response is most significantly affected in the dawn and dusk sectors, where the contribution of the partial ring current to the local magnetic record is most pronounced.

Keywords: Regional Space weather, Magnetic fluctuations, Prompt-Penetration Electric Fields

1 Introduction

A geomagnetic storm (GS) is a temporary disturbance of Earth’s magnetosphere caused by the injection of energy and charged particles into the system, leading to a significant reduction in the geomagnetic field intensity. These GS are typically triggered by solar phenomena such as Coronal Mass Ejections (CMEs) or high-speed solar wind (HSSW) streams. Due to their potential impacts on technological systems—including disruptions to communications, navigation, and power grids—GSs are among the most extensively studied phenomena in space weather research.

During the Main Phase (MP) of a GS, energy and charged particles are transported into Earth’s inner magnetosphere. Since the magnetosphere is dynamically coupled with the ionosphere, particularly at high latitudes [C.T Russell, 2016], the ionosphere exhibits pronounced responses to these perturbations. One critical ionospheric effect is the generation of storm-time electric fields, which can be manifested at different latitudes:

- Prompt Penetration Electric Fields (PPEF) [Nishida, 1968a,b]
- Disturbed Dynamo Electric Fields (DDEF) [Blanc and Richmond, 1980]

The concept of PPEF was first introduced by Nishida [1968a,b], who observed periodic magnetic fluctuations between 4 hours and 30 minutes in ground magnetometers located around the auroral oval and the dip equator, both showing nearly identical fluctuations. Later, Vasyliunas [1970], Jaggi and Wolf [1973] developed the first model explaining the penetration of convective electric fields ($\mathbf{E}_C$) during disturbed periods, propagating through Region 1 Field-Aligned Currents (R1-FAC). During quiet periods, $\mathbf{E}_C$ is shielded by a polarized electric field induced by Region 2 Field-Aligned Currents (R2-FAC). However, during the MP of a GS, magnetic reconnection leads to an almost immediate intensification of R1-FAC activity due to the injection of solar wind particles and energy. Since R2-FAC depends on ring current activity to strengthen, it initially fails the shielding of $\mathbf{E}_C$, allowing its transfer to the ionosphere through R1-FAC. This process continues until R2-FAC intensifies its activity though the intensification of ring current, allowing R2-FAC balancing R1-FAC activity, which can take from tens of minutes to several hours [Wei et al., 2015]. During this period of imbalance, $\mathbf{E}_C$ penetrates into the mid- and low-latitude

ionosphere, being that the reason to be called Penetration Electric fields. Furthermore, Kelley et al. [1979], Wei et al. [2015] showed that PPEF depends on the southward orientation of the interplanetary magnetic field (IMF) component B_Z and can also be triggered by sharp directional changes in B_Z .

DDEF on the other hand, are associated with an ionospheric current driven by the injection of charged particles into the auroral ionosphere. This injection, produces a Joule heating and leads to the circulation of equatorward thermospheric winds which drag charged particles (protons) Blanc and Richmond [1980]. With the dragged protons, it gets to a charge separation which induces a Pedersen current in the same direction of neutral wind flow. Near the equator, the Coriolis effect produces proton accumulation and westward circulation occur around the dip equator, generating a poleward electric field. Consequently, a second Pedersen current is induced in the poleward direction, opposing the first current. At mid-latitudes ($\sim 40^\circ$), the above commented phenomena, drives Hall currents, which are interrupted at the terminators (limit of day-side with night-side) and get a closure or circuit at adjacent latitudes. This process results in vortex-shaped currents known as the Disturbed Dynamo, characterized by directions opposite to the Solar Quiet (SQ) current. For this reason, the disturbed dynamo is also referred to as the anti-SQ current.

At mid-low latitudes ($20 - 30^\circ$), geomagnetic storms can significantly affect electron density through two primary mechanisms: Prompt Penetration Electric Fields (PPEF) and disturbance dynamo-driven neutral winds. On the dayside, an eastward PPEF causes an upward $\mathbf{E} \times \mathbf{B}$ drift, lifting the plasma to higher altitudes where recombination is slower, often leading to a positive ionospheric storm (TEC increase). In contrast, the storm-induced circulation from high to low latitudes transports molecular-rich air ($[O_2]$, $[N_2]$), increasing the $\frac{[N_2]}{[O]}$ ratio. This change in atmospheric composition enhances the loss rate of atomic oxygen ions (O^+) through recombination, which can lead to a negative ionospheric storm (TEC decrease) [Vijaya Lekshmi et al., 2011, Prölss, 1995].

Both PPEF and DDEF were studied by Fejer et al. [1983], who observed their effects using incoherent scatter radar at Jicamarca latitude. Later, Amory-Mazaudier [1982] detected DDEF above Saint-Santin latitude, with results consistent with the disturbed dynamo model. On the other hand, Le Huy and Amory-Mazaudier [2005] modeled by the first time the magnetic field associated with ionospheric disturbances, assuming that, for mid-low latitudes, they are mainly driven by DDEF and PPEF. They considered that during the MP, PPEF magnetic effects are dominant, whereas during the recovery phase (RP), regional magnetic field variations are mainly driven by DDEF. Following this initial approach, several studies [Le Huy and Amory-Mazaudier, 2008, Zaka et al., 2009] have focused on analyzing local responses at different latitudes and longitudes to analyze the different local responses associate to DDEF phenomena. In Liu et al. [2013], a statistical study was conducted to understand ionospheric responses as a function of local sectors during GS onset. Later, Younas et al. [2020] proposed using Fourier analysis to isolate magnetic fluctuations related to DDEF and PPEF based on their quasi-periodic behavior, identifying periods of approximately 24 hours for DDEF and less than 4 hours for PPEF.

In case of Mexico, regional space weather has a relatively recent history. On 2014, it was form the Mexican Space Weather Service (SCIESMEX), leading to combined efforts to understand the regional space weather phenomena, in the middle of a global perspective Gonzalez-Esparza et al. [2017]. From this teamwork, studies like Corona-Romero et al. [2018, 2017] has been focused on local geomagnetic response during GS periods, and the computing of a local K index specific for Mexico. On the other hand, Romero-Hernandez et al. [2017], Sergeeva et al. [2017], Sergeeva et al. [2020] found important ionospheric activity related to geomagnetic storms. In Castellanos-Velazco et al. [2024], the main focus was analyzing the local geomagnetic response related to PPEF and DDEF phenomena, during 20 strong GS, from 2003 to 2018, where one of the main conclusions were the need of comparing the geomagnetic response in the center of Mexico with different locations at similar geomagnetic latitudes.

In this paper, we analyze the PPEF magnetic signatures at different longitude sectors, looking for the sector where these fluctuations are more intense. We compare the results with the Asymmetric Ring Current activity in order to analyze it's influence on PPEF strength and so, on local geomagnetic response. We selected the GS of march 17th, 2015 which is also known as the St. Patrick's event. This GS is known to be the most intense GS during the solar cycle 24th, reaching -274 nT during its MP.

2 St. Patrick Geomagnetic Storm

The geomagnetic storm (GS) analyzed in this work is the intense event that occurred on March 17, 2015, widely known as the St. Patrick's Day storm. It is considered the most intense storm of solar cycle 24 Wu et al. [2016]. The storm was triggered by a coronal mass ejection (CME) associated with a C9.1 solar flare and a subsequent dimming observed on March 15. This CME, recorded as a partial halo by the SOHO/LASCO coronagraphs, had an estimated speed of approximately 660 km/s.

Figure 1 provides an overview of the interplanetary conditions that led to the St. Patrick's Day storm. A more detailed description about the trigger event is provided by Wu et al. [2016]. Panel (a) shows the total interplanetary magnetic field, B_T (black line), and its north-south component, B_Z (red line). Panels (b) and (c) presents solar wind properties, including the induced Interplanetary Electric Field (E_Y), calculated as $-V_X \times B_Z$; and dynamic pressure. Finally, panel (e) displays the geomagnetic response through the SYMH and ASYH indices (green and yellow lines respectively).

The key milestones of the event are marked by three vertical dashed lines in Figure 1. The first line (left) indicates the arrival of an interplanetary shock (IPS) at 04:30 UT at the L1 point. The IPS signature is clear from the step-like increases in B_T , E_Y and P_{dyn} . This shock compressed the magnetosphere, causing a Sudden Storm Commencement (SSC) visible as a sharp increase in the SYMH index between 04:30 and 07:00 UT, highlighted by light gray shading.

The region following the shock, the CME sheath, is identified by highly fluctuating B_T and B_Z , enhanced P_{dyn} . A period of southward B_Z between 06:00 and 09:00 UT initiated the what we called, the first stage of main phase (MP), driving the SYMH index down to its first minimum of -80 nT.

The second dashed line (14:00 UT) marks the arrival of the magnetic cloud, identified by Wu et al. [2016] as the main driver of the GS. Its passage is identified by a smooth, strong rotation of the magnetic field, a drop in temperature, and so, a corresponding low plasma beta. A sustained and intense southward B_Z component (reaching -40 nT) within the cloud marked the beginning of the second (and more intense) stage of the MP. The SYMH index decreased sharply, reaching its global minimum of -234 nT at 22:47 UT. The entire main phase period is shaded in dark gray in panel (e).

The third dashed line (23:00 UT) indicates the trailing edge of the magnetic cloud and the entry into a high-speed solar wind (HSSW) stream. While the high solar wind speed persists, the geomagnetic activity subsides as B_Z turns northward and the E_Y drops close to zero, ending the geoeffective driving of the storm.

Since we are interested in the effects of the PRC during its highest activity altogether with PPEF effects, we will focusing on a time window of a few hours during MP. This time window is going to be in function of periods when $E_Y > 0.5 mVm^{-1}$ and $P_{dyn} > 29.71 nPa$. The first criteria is based on the discussion in Burton et al. [1975], where they find the energy injection during MP is dependent of E_Y , setting $E_{Y0} = 0.5 mVm^{-1}$ as a minimum value. On the other hand, Xie et al. [2008] set $P_{dyn} > 15 nPa$ as a good estimator for a geoeffective dynamic pressure for strong geomagnetic storms. Considering that 15 nPa is a very low threshold for the period between march 15 - 17 as it can be seen in the Figure, we set the threshold through computing the 95% of Cumulative Density Function (CDF). Thus, we obtained a threshold of 29.71 nPa. Hence, obtain two possible time windows: 6-7 UT and 13-18 UT. Since the second period corresponds to the highest ASYH values, we selected this period for study, and its highlighted by the violet shading. Hence, during the selected time window, we expect higher injection and Electric Field penetration (under shielding) and PRC activity (Over shielding), which implies higher related magnetic fluctuations.

3 Procedure

3.1 Geomagnetic Data

The geomagnetic data analyzed in this paper were downloaded from the INTERMAGNET public database (*International Real-Time Magnetic Observatory Network* [INTERMAGNET, 2021]), corresponding

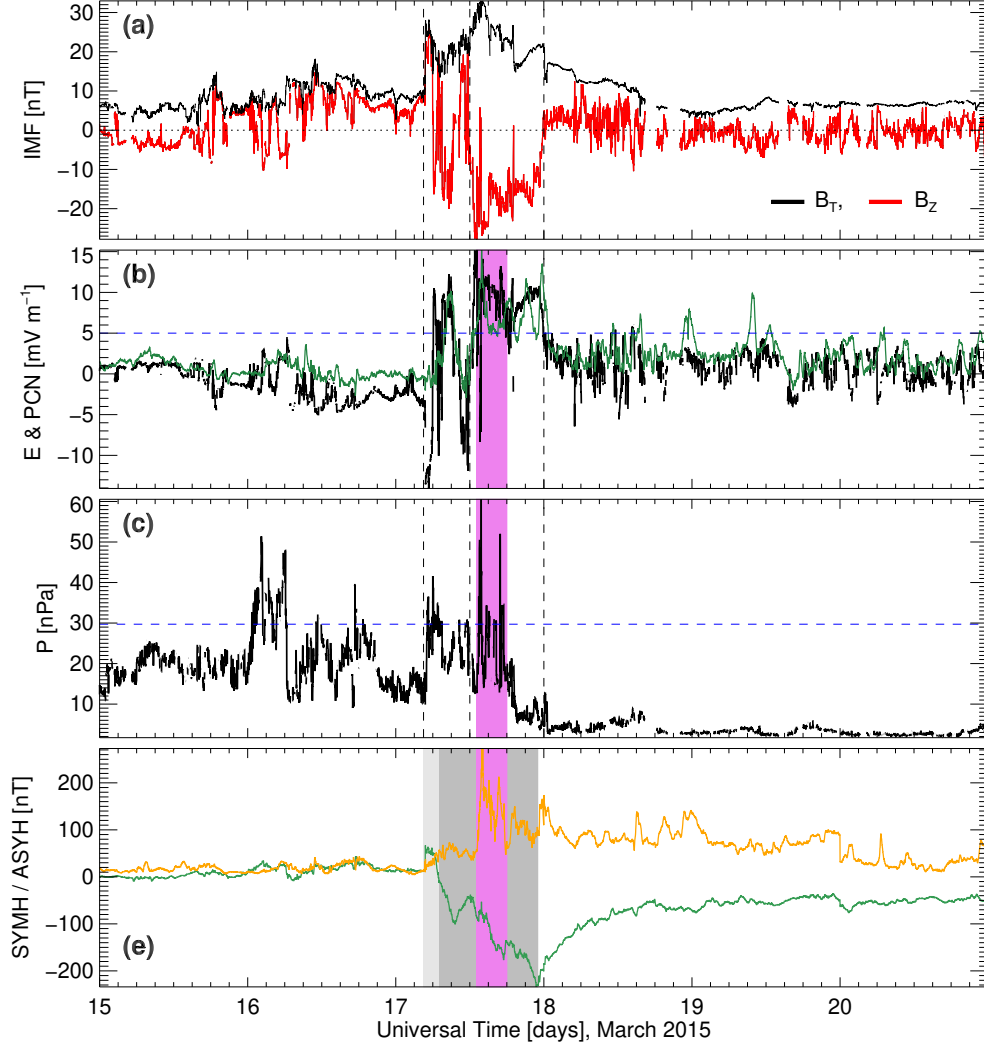


Figure 1: Interplanetary measurements. Panel (a): interplanetary magnetic field total component (black line) and z component (red line). Panel (b): E_Y Interplanetary Electric Field (blue line). Panel (c): Dynamyc Pressure [nPa]. Panel (d): SYMH (green line) and ASYMH index (orange line), highlighting the initial phase (light gray shade), main phase 1 (medium gray shade) and main phase 2 (dark gray shade).

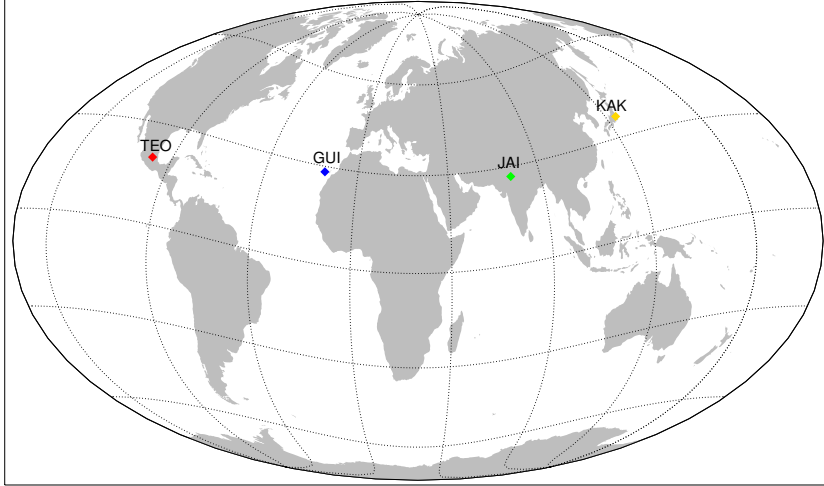


Figure 2: Geographic position of the four magnetic OB, distributed at $70 - 90^\circ$ from each other

to the observatories (OB) listed in Table 1.

Table 1: List of observatories considered for this study. Each OB is separated longitudinally at $70 - 90^\circ$ apart. The OB selected for the main analysis are light gray shaded.

Observatory	IAGA code	Country/ Territory	Latitude		Longitude	
			Geographic	Magnetic	Geographic	Magnetic
Beijing	BMT	China	40.3 N	30.53 N	116.2 E	172 W
Cheongyang	CYG	South Korea	36.37 N	27 N	126.854 E	162.43 W
Guimar	GUI	Spain	28.317 N	33.21 N	16.433 W	61.1 E
Honolulu	HON	USA	21.317 N	21.49 N	158 W	90.03 W
Jaipur	JAI	India	26.917 N	18.42 N	75.8 E	150.47 E
Kakioka	KAK	Japan	36.232 N	27.8 N	140.186 E	149.73 W
Lanzhou	LZH	China	36.083 N	26.23 N	103.833 E	176.79 E
San Juan	SJG	USA, Puerto Rico	18.11 N	28.19 N	66.15 W	6.1 E
Tamanraset	TAM	Argelia	22.783 N	24.3 N	5.571 E	82.29 E
Teoloyucan	TEO	Mexico	19.747 N	27.81 N	99.182 W	28.4 W

To study the different geomagnetic responses, we focused on data from observatories (OB) listed in Table 1. As it will be seen later in Section B, OB which are longitudinally too close, will have similar results. Thus, we focus in the four OB highlighted in the Table, which are longitudinally separated by $60 - 90^\circ$ from each other and are highlighted in Figure 1. As mentioned before, the analysis will be focused during a period of maximum injection, defined by a large E_Y interplanetary electric field penetration and high dynamic pressure and maximum activity from the partial ring current (PRC), measured by ASYH index (273 nT). Both mechanisms imply the under shielding and overshielding of the $PPEF$, thus we could expect high H_{PPEF} fluctuations during this period, from 14:07 to 19:00 UT (5 hours) and shortly after, from 19:00 UT to 23:00 UT which coincides with a period of high magnitude E_Y but low P_{dyn} (*approx 6 nPa*). Interestingly, the period TW also coincides with each one of the selected OB being transitioning through the four local time sectors:

- Dawn-Noon sector (TEO), 06:01 - 12:00 LT.
- Noon-Dusk sector (GUI), 12:01 - 18:00 LT.
- Dusk-Midnight sector (JAI), 18:01 - 00:00 LT.
- Midnight-Dawn sector (KAK), 00:01 - 06:00 LT.

3.2 Magnetic fluctuations of ionospheric origin

It is well accepted that [Rishbeth and Garriott, 1969, Knecht and B.M, 1985, Gjerloev, 2012, van de Kamp, 2013] local recordings H_{loc} of a ground magnetometer can be considered a linear combination of different geomagnetic sources:

$$H_{loc} = H_{SQ} + H_0 + H_{MT} + H_{others} + H_I, \quad (1)$$

where H_{SQ} represents the diurnal variation and H_0 is the main field during a month window. H_{MT} represents the variations induced by the intensification of magnetospheric currents during the GS. In the case of H_{others} , it corresponds to magnetic sources less significant, like (and no limited to)) magnetotellurics, crustal magnetization, etc; which can be neglected for not being considered as relevant compared to the others mentioned. Finally, H_I Le Huy and Amory-Mazaudier [2005] corresponds to the magnetic variations driven by ionospheric current activity during the GS. Thus, the local geomagnetic response during a GS is given by:

$$\Delta H = H_{loc} - (H_{SQ} + H_0). \quad (2)$$

We provide a detailed explanation of how we model the diurnal variation and the monthly baseline in Appendix A.

From Le Huy and Amory-Mazaudier [2005], Amory-Mazaudier et al. [2017], Younas et al. [2020], for mid-low magnetic latitudes, with λ as the geomagnetic latitude of each OB, we can roughly approximate H_{MT} as $H_{MT} \approx SYMH \cdot \cos(\lambda)$. Thus, H_I can be approximated as:

$$H_I = H_{DDEF} + H_{PPEF} + H_{others} \approx H - (H_{SQ} + H_0 + SYM - H \cdot \cos(\lambda)), \quad (3)$$

where H_{DDEF} and H_{PPEF} represent the magnetic fields associated with the disturbed dynamo electric fields and the prompt penetration electric fields, respectively. H_{others} on the other hand remains as a noise.

In order to isolate H_{DDEF} and H_{PPEF} from each other, we proceed under the assumption that both ionospheric currents drive quasi-periodic fluctuations [Nishida, 1968a, Blanc and Richmond, 1980, Younas et al., 2020]. Hence, through a Fourier analysis, we apply pass-band filters for frequencies between $69.5 - 550\mu\text{Hz}$ (corresponding to periods in 4 hours to 30 minutes range).

Given the role of PRC within local geomagnetic response, Younas et al. [2020] applies an extra step, where:

$$H'_I = H_I - ASYH \cdot \cos(\lambda), \quad (4)$$

in order to remove also the contribution from ASYH index. This suggest that PRC effects remains within H_I , specially in the night side and early morning [Takahashi et al., 1991]. However, given the short duration of PRC activity, in comparison with the symmetric part of the system, combined with the limited longitudinal extension of the PRC would mean that Eq 4 shouldn't be applied equally for the OB, specially when this OB are located by $60 - 90^\circ$ degrees from each other.

Hence, the analysis will be mainly focused on comparing the result of H_I with ASYH index. Given the order of magnitude of ASYH, compared with H_{PPEF} (and H_{DDEF}), this would mean that ASYH will be the main source for H_I , depending on the OB position with respect to local Time.

4 Results

In the following section, we focus on the results related to the geomagnetic response to the St. Patrick's GS for each OB listed in Table 1. Then we show the results of Equation 3 for isolating H_I , and an analysis related to local time sector in which the GS developed as well as a comparison between each OB result.

In Figure 3, we show in top panel, the ASYH index, and the computed ionospheric magnetic contribution H_I (red lines) for GUI, JAI, KAK and KAK (second to fifth panels). The dashed vertical

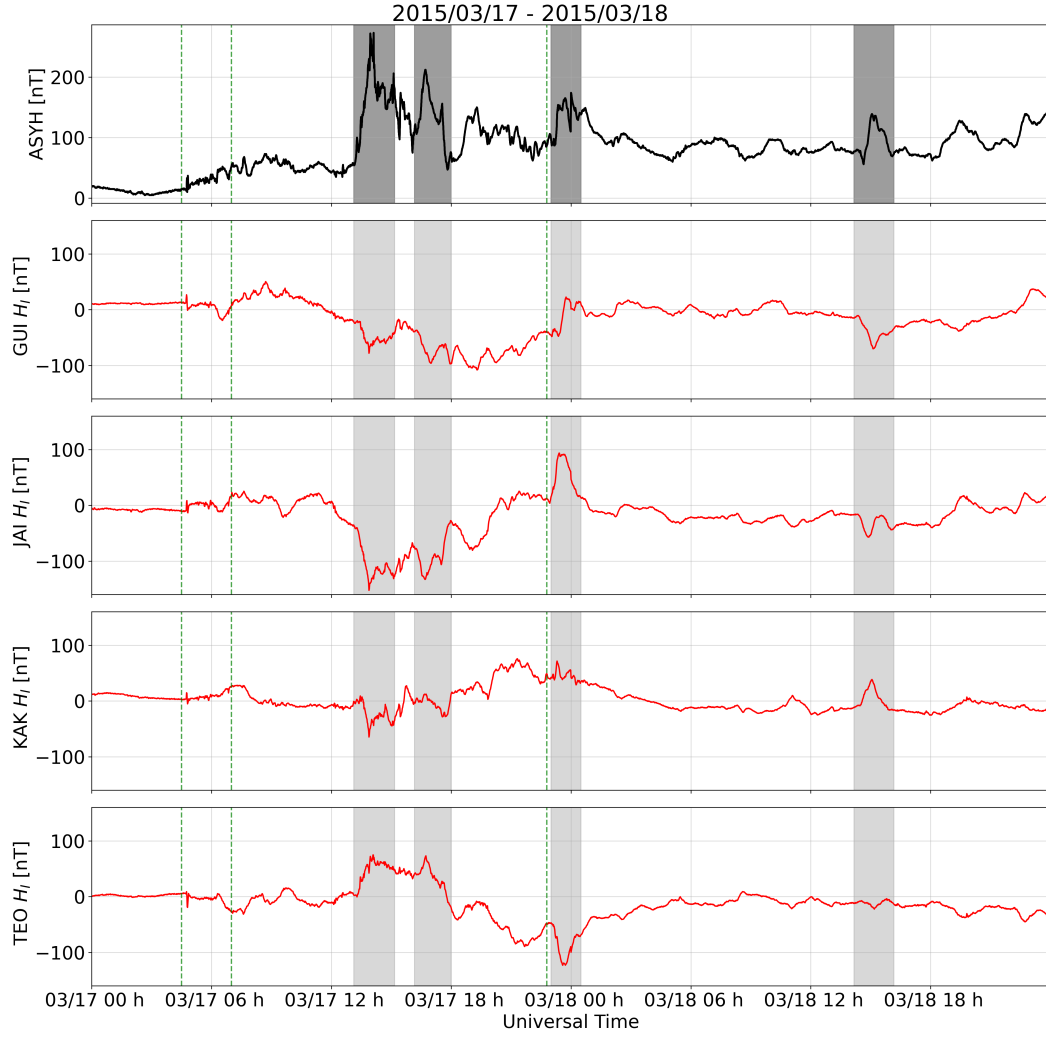


Figure 3: Comparison between ASYH index (top panel) with H_I from Eq 3, downward: GUI, JAI, KAK and TEO. The green dashed vertical lines represent the beginning of SSC, MP and RP.

lines represent the transition of quietness-SSC, SSC-MP and MP-RP. Our main attention will be on the first 2 days after the onset of the GS, march 17-18 which envelops the MP and 1 day of RP.

An easy way to acknowledge periods when H_I presents an important contribution from PRC is regarding weather or not it shows a similar/opposed pattern to ASYH index. This is mostly seen during and after TW. In the Figure 3, notice how in the highlighted periods (like 13:00 - 15:00, 16:00 - 18:00, 23:00 - 00:30 UT and 14:00 - 16:00 UT on the next day), H_I present a behavior which can be similar(opposed) to ASYH index, depending on the OB. From Makarov [2022], it is implied that ASYH and SYMH indices are anti-correlated. Thus, H_I behaving opposed(similar) to ASYH means that PRC might be contributing to an intensification (attenuation) to their respective local geomagnetic response ΔH . Among the results, JAI H_I is the one with higher intensities during the highlighted periods, followed by TEO (in opposed way).

The different intensities in H_I OB results (specially during the highlighted periods) suggest how PRC may impact differently depending on the local time sector (being JAI close to dusk, TEO close to dawn, GUI close to noon and KAK close to midnight during the first highlighted period).

To quantitatively confirm weather PRC contribution remains in ΔH and so, presents a noise contribution factor within H_I . In Figure 4, we analyze the check on the Pearson correlation between H_I and ASYH during TW (13:00-19:00 UT), which is the period when it is registered the highest Injection process and highest PRC activity occurs.

As mentioned before, strongly positive or negative ρ coefficient can be interpreted as ΔH being influenced (attenuated or intensified) by PRC activity. The two observatories showing stronger correlation (or anti-correlation) are JAI and TEO which coincides with these OB being closer to the terminators during the period of highest PRC activity.

A wider perspective of the ASYH - H_I correlation is shown in Figure 5

5 Discussion

Following with the influence of PRC on H_{PPEF} fluctuation amplitudes, we carried out an analysis of H_{PPEF} distribution magnitudes, from march 17th to march 18th, shown in Figure 6. From top panels, we observe that H_{PPEF} magnitudes (blue bars) in each OB seems to approach to a Gaussian distribution (fitted with a red line). All OB present media $\mu \sim 0$ nT and a very small skewness, seen at first glance, being JAI the farther from 0 ($\mu = 0.67$). The standard deviation σ seen in each OB has a maximum difference goes from 5.2 to 8.15, being not so significant.

Both, the almost symmetry of the distribution and the center near zero are consistent with H_{PPEF} having a periodic behavior. The shape of the distribution on the other hand, so as the length of both tails are a good indicator of the intensity of these fluctuations. From here, it can be seen that the longest tails corresponds to JAI and TEO which is consistent with JAI and TEO having higher peaks of H_{PPEF} amplitudes. GUI and KAK on the other hand, present greatest and minor dispersion respectively. In case of GUI, this is consistent with H_{PPEF} amplitudes not having a noticeable contrast during TW, meanwhile KAK presents minor activity in general, but with a higher contrast of amplitudes during TW.

From here, we are interested in estimating the point at which H_{PPEF} can be considered as extreme, for each OB. We refer this point as *Threshold*. These thresholds can be used for determining periods of more significant PPEF activity. Thus, the higher the threshold, higher PPEF activity in a certain region.

First, from the good symmetry in all distributions, we can consider the use of an absolute value threshold. From this approach, we set each threshold as the 90% of each cumulative H_{PPEF} probability distribution function, shown in bottom panels of Figure 6 (blue lines). The threshold is indicated by the red dashed line for the 2 days period.

The highest thresholds correspond to JAI (28.63 and 40.98 nT) followed by TEO (20.31 and 33.79), meanwhile, in GUI and KAK, have significantly minor thresholds. What this difference may indicate is how H_{PPEF} intensity is affected by the LT sector at the moment of the highest ASYH activity. This result is consistent with the results of $\rho_{1,2}$ for these OB, which might reinforce the idea of H_{PPEF} being influenced by PRC activity, indicating not just the impact of PRC activity but also on the proximity of

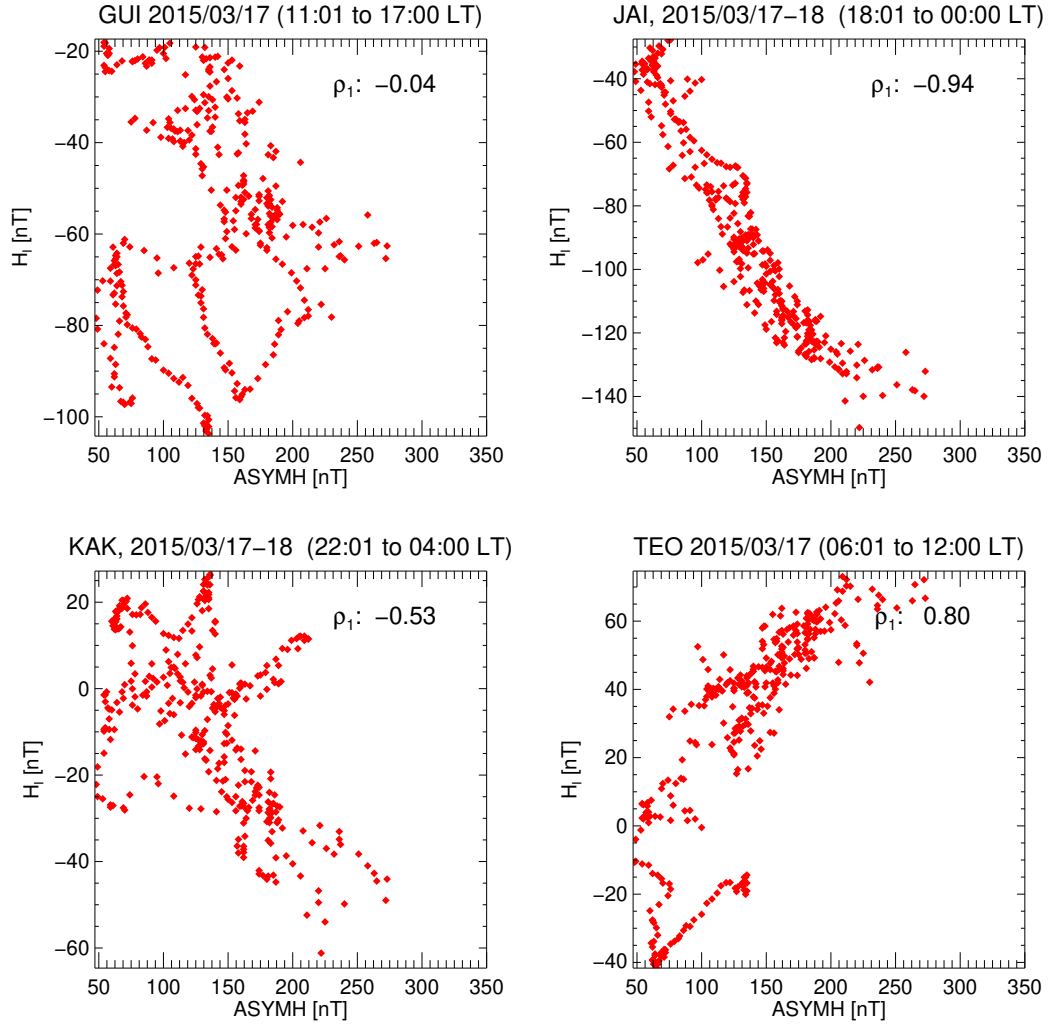


Figure 4: Correlation Analysis ASYH and H_I , during the period 1300 - 1900 UT, which is consistent with the period of highest ASYH activity

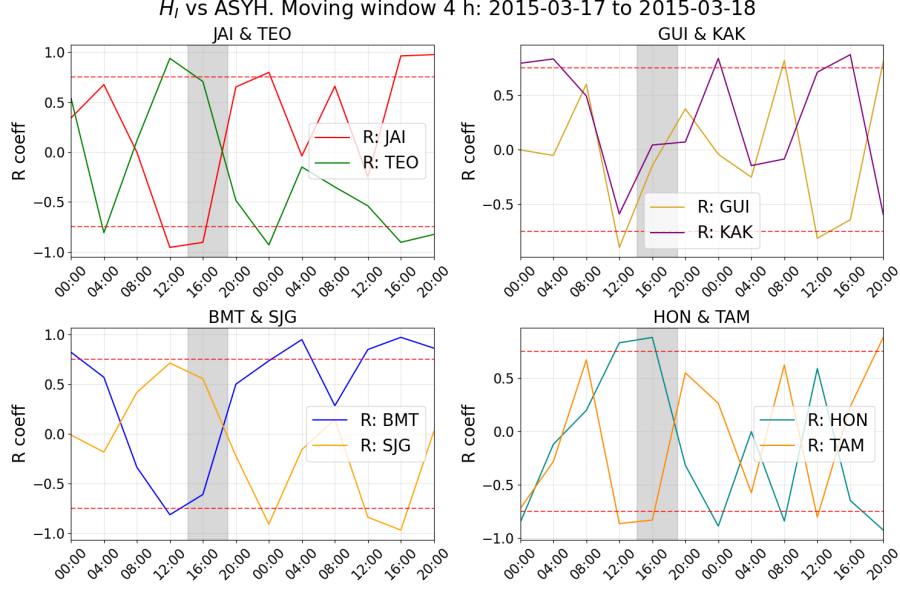


Figure 5: Correlation Analysis ASYH and H_I , during the period 1300 - 1900 UT, which is consistent with the period of highest ASYH activity

JAI and TEO to the dusk-midnight sectors during the highest PRC activity.

6 conclusions

Within this project, we analyzed longitudinal magnetic fluctuations driven by Prompt Penetration Electric Fields (PPEF) during the intense St. Patrick's Day geomagnetic storm on March 17, 2015. To achieve this, we utilized geomagnetic data from ground-based observatories separated by 60–90° in longitude.

A Processing magnetic data

It is crucial to eliminate regular variations from magnetic records to isolate magnetic contributions of interest. The main field H_0 can be removed by modeling a monthly baseline. This baseline was approximated by computing an array of daily mode values for March 2015, following Gjerloev [2012]. These daily mode values were obtained from data between 00:00 and 03:00 LT ($H_{0:3}$), then H_0 was computed by a linear least squares fitting from these daily values. To exclude data from disturbed days for the fitting, we adapted the procedure outlined in van de Kamp [2013]. Specifically, we applied a moving hourly window to calculate the hourly interquartile range (IQR). Disturbed days were identified by setting a variation threshold where:

$$IQR_{med}^i > threshold = disturbed \quad (5)$$

Here, IQR_{med}^i is the median of each daily IQR values, so if more than 50% of IQR values for a given day, surpass a determined threshold, then that day is considered as disturbed day and depreciated for the LS fitting.

The threshold was set, based on the criteria seen in Solari et al. [2017]. Basically, we analyze the monthly distribution of maximum IQR values (IQR_i^{max}) within tri-hours IQR bins, by shifting another moving window and sorting those peaks, so $IQR_1^{max} < IQR_2^{max} < \dots < IQR_n^{max}$, in order to get a cumulative IQR^{max} distribution.

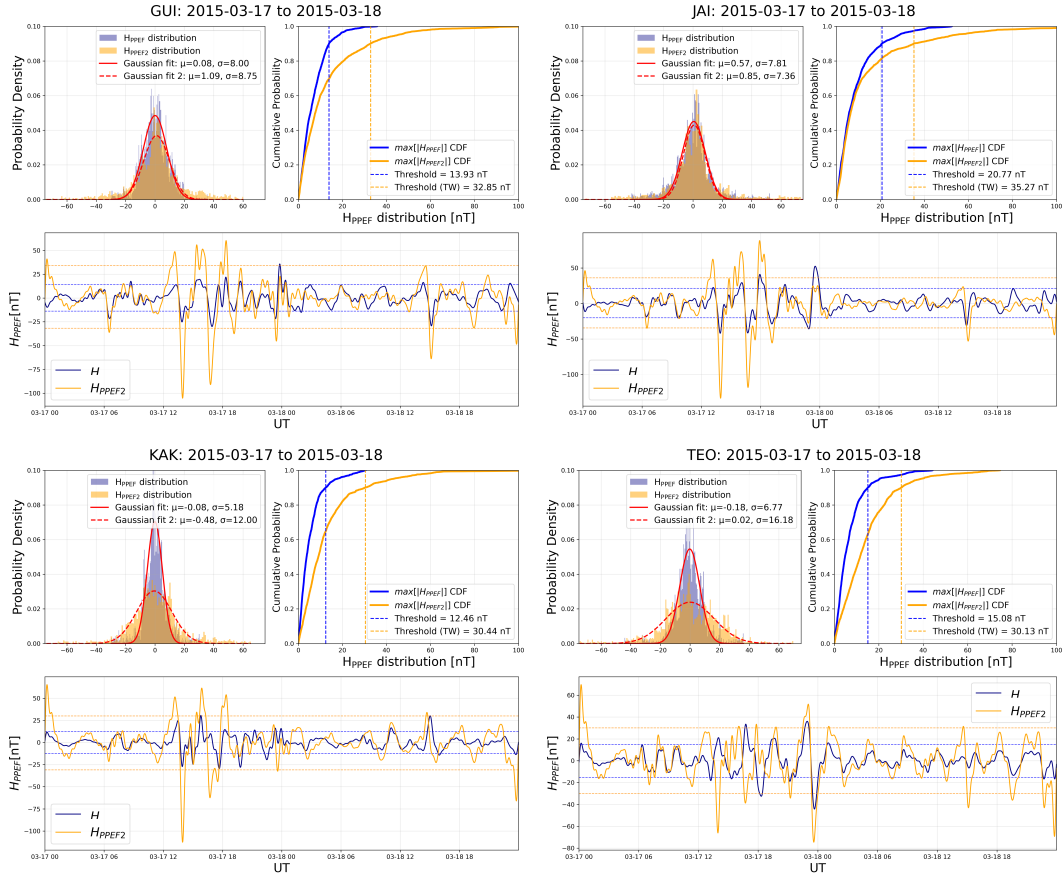


Figure 6: H_{PPEF} distribution for each OB. Color dashed vertical lines represent the a threshold for H_{PPEF} to be considered as intense amplitudes

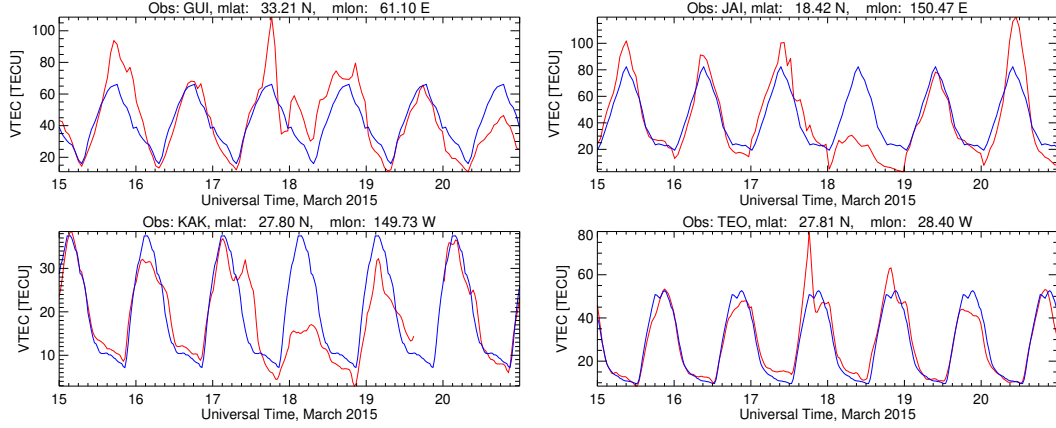


Figure 7: Total Electron Content from LPAL (GUI), NPGJ (JAI), TSK2 (KAK) and UCOE (TEO)

Hence, we have to compute the parameters of the General Pareto Distribution (GPD) Pickands [1975] which is a type of distribution used for modeling other distributions extreme values, and it is expressed as:

$$f(x) = \begin{cases} \frac{1}{\sigma} \left(1 + k \frac{x-u}{\sigma}\right)^{-1-\frac{1}{k}} & \text{if } k \neq 0 \\ \frac{1}{\sigma} \exp\left(-\frac{x-u}{\sigma}\right) & \text{if } k = 0 \end{cases} \quad (6)$$

where the three GPD parameters are shape k , scale σ and location u for the cumulative IQR_i^{max} . The last parameter, u is the one which can be considered as a threshold point [Hosking, 1990, Solari et al., 2017]. Then, for each IQR_i^{max} , we compute $[k_i, \sigma_i, u_i]$, then we apply a modified Anderson-Darling test. The threshold, u_t , is considered once the test gives the most optimum validation result.

To compute the diurnal variation, we identified what we refer to as local quiet days (LQD) for each station. Following the methodology in van de Kamp [2013], we considered the five days with the smallest daily IQR peaks as LQDs. We then computed the average of the H component data during these five quietest days:

$$\langle LDQ_H(t) \rangle = \frac{1}{5} \sum_{i=1}^5 H_i(t) - \langle H_i 0 : 3 \rangle \quad (7)$$

where we normalize the offset of each daily data by subtracting the average of the first four hours of magnetic data in local time, as indicated by the second term in Equation 7. Subsequently, we model the diurnal variation H_{SQ} by applying a low-pass filter to the Fourier Transform of the result from Equation 7, following the procedure in Equation (1) of van de Kamp [2013].

B ionospheric and magnetic response

B.1 Total Electron Content

The overall ionosphere dynamics in each considered region were characterized by the vertical Total Electron Content (TEC) variations in the vicinity of the chosen magnetic OB. TEC values were derived from GNSS measurements using TayAbsTEC method Yasyukevich et al. [2015a,b], Mylnikova et al. [2015]. In particular, data from UCOE, LPAL, NPGJ and TSK2 receivers was used to obtain TEC values for TEO, GUI, JAI and KAK regions, respectively.

In Figure 7, it is shown the TEC responses (red line) during the period of the GS, compared with the median of the 27 days TEC prior to the GS (blue line). However, given the different location of the GNSS stations, it is usual to derive different expected daily ranges (in GUI/LPAL is around 16-70 TECu; JAI/NPGJ is 20 - 80 TECu; KAK/TSK2 is 7 - 38; TEO/UCOE is 10 - 55 TECu). The same goes for different amplitude responses over/bellow the median, so checking only the TECu magnitude may not be possible to compare the results between the different GNSS stations.

Another issue is that we have to consider that the open data obtained for this work, doesn't have the best quality, even more considering the period of time. The quality of data is a sum of errors which depend on the satellite swarm, so as the receptor itself. We could smooth some of these errors but couldn't be avoided at all. Considering also that we are studying results from points in different longitudinal sectors (Mexico, Canarias Islands, India and Japan), there can be issues by comparing absolute TEC values. To deal with this issue, we instead analyze relative values, defined by:

$$\Delta TEC = \frac{TEC_{obs} - TEC_e}{TEC_e}, \quad (8)$$

where TEC_{obs} is the observed response during the period of interest and TEC_e is the expected TEC, derived from the 27 days median TEC. ΔTEC is then, measuring TEC percentage difference between the expected value and the TEC observed during the GS Vijaya Lekshmi et al. [2011], Sergeeva et al. [2017].

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